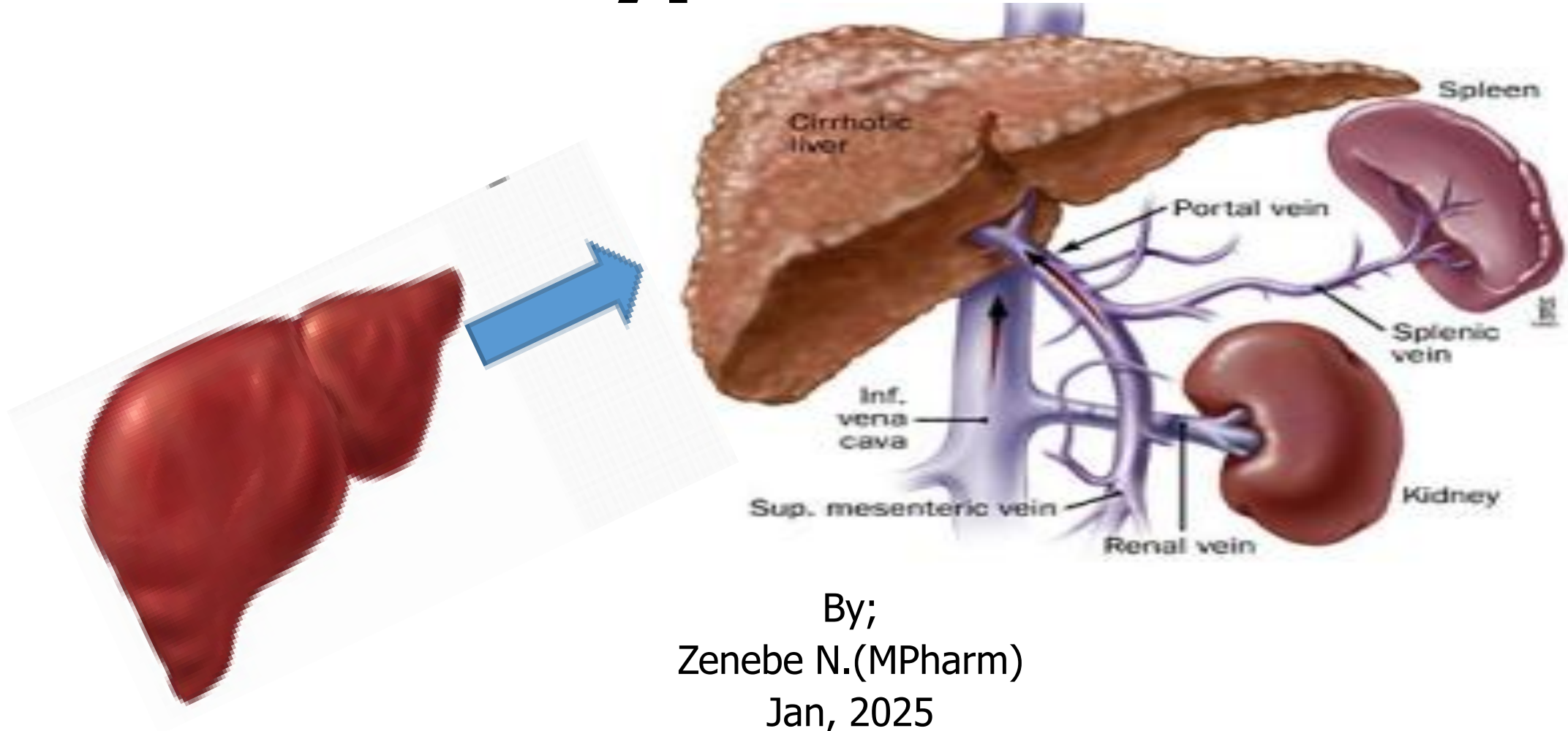


Cirrhosis and Portal Hypertension



By;
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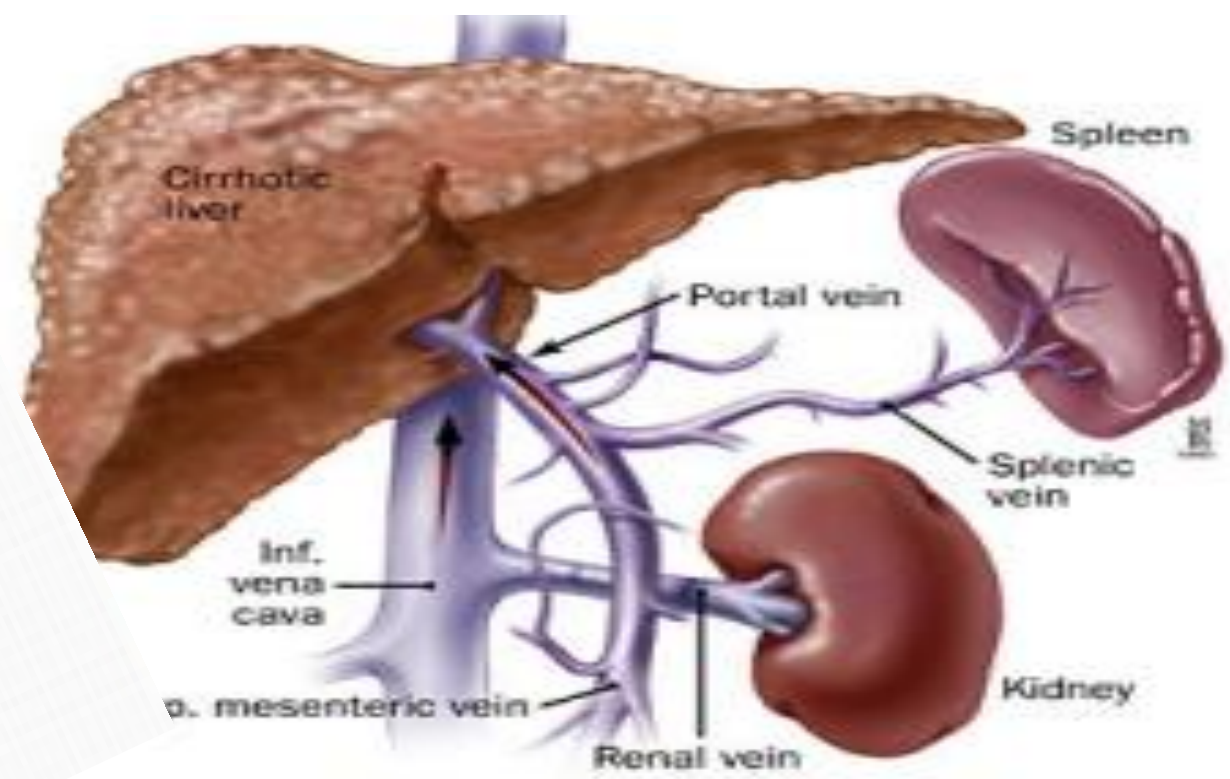
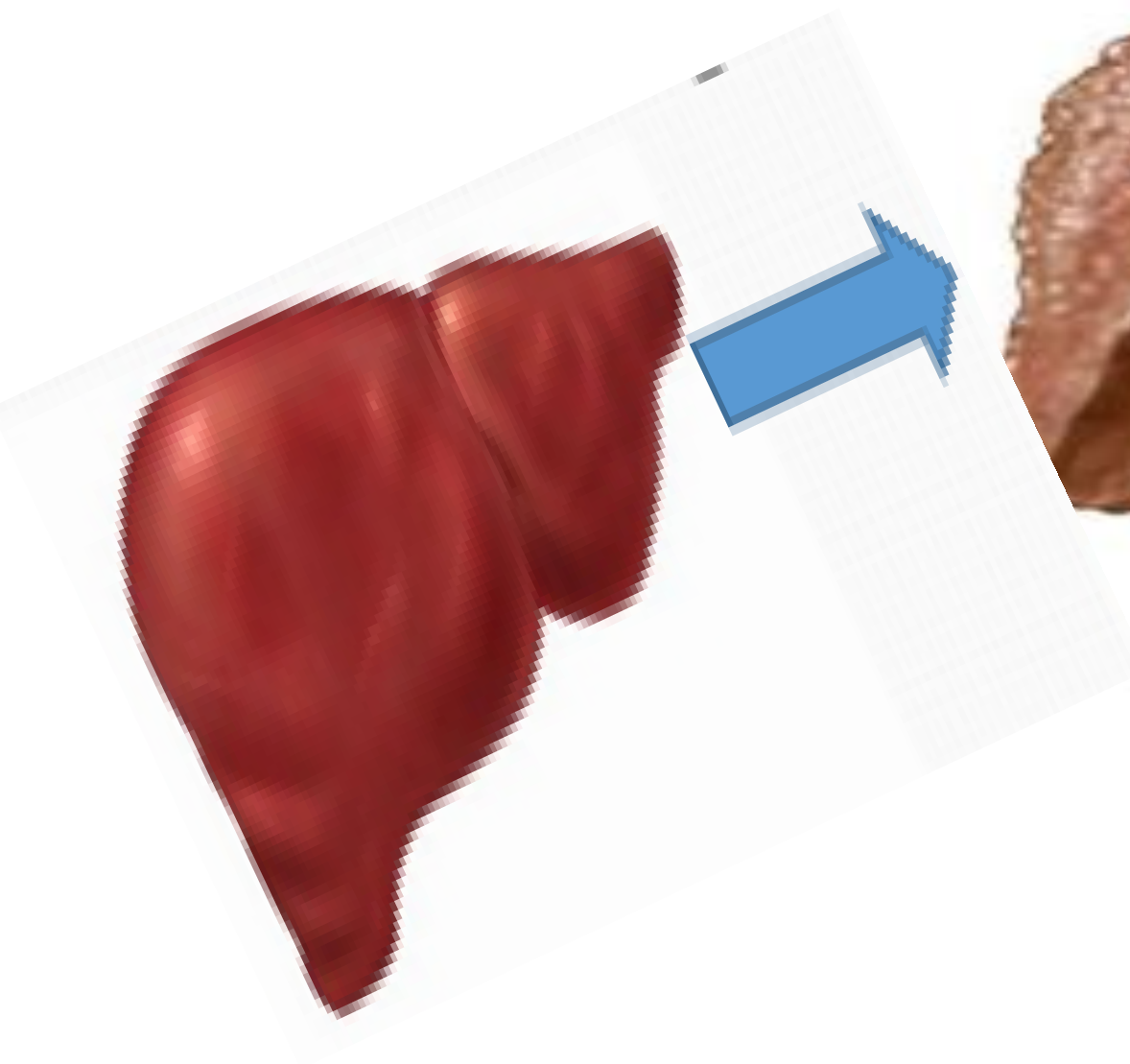
Learning Objectives

❑ **Upon completion of this chapter, students will be able to:**

1. Explain the epidemiology, etiology, and pathophysiology of cirrhosis.
2. Identify the potential complications associated with cirrhosis.
3. Recognize the typical clinical presentation and diagnostic features of a patient with cirrhosis.
4. Define portal hypertension and describe its pathophysiology and associated complications.
5. Develop a treatment plan for the primary and secondary prophylaxis of variceal bleeding.
6. Formulate evidence-based strategies for managing acute variceal bleeding in cirrhotic patients.
7. Describe the clinical features of ascites and its relationship to spontaneous bacterial peritonitis (SBP).
8. Design a comprehensive management plan for a patient with ascites, including preventive and therapeutic interventions.
9. Recognize clinical scenarios and Recommend an evidence-based antibiotic regimen for patients with SBP
10. Define hepatic encephalopathy (HE), detailing its precipitating factors and various clinical manifestations.
11. Develop effective treatment recommendations for patients with HE.

Cirrhosis

- Cirrhosis is a severe, chronic, potentially irreversible disease associated with significant morbidity and mortality
- Cirrhosis is an advanced state of liver fibrosis.
- Fibrosis, the replacement of injured tissue by scar tissue,
 - distortion of the hepatic vasculature leading to shunting of hepatic blood supply.
- Chronic liver injury → → damage to normal liver tissue → → the development of **regenerative nodules** surrounded by dense fibrotic material,
 - diagnostic hallmarks of cirrhosis.
- The distorted architecture of the cirrhotic liver **impedes portal blood flow, interferes with hepatocyte perfusion, and disrupts hepatic synthetic functions such as the production of albumin.**



- Clinical consequences of cirrhosis include
 - ✓ Increased intrahepatic resistance leading to portal hypertension, varices, and variceal bleeding;
 - ✓ Ascites
 - ✓ Infection
 - ✓ Encephalopathy
 - ✓ Hepatocellular carcinoma
- Additionally, when advanced, cirrhosis can also lead to the development of both renal and pulmonary dysfunction.

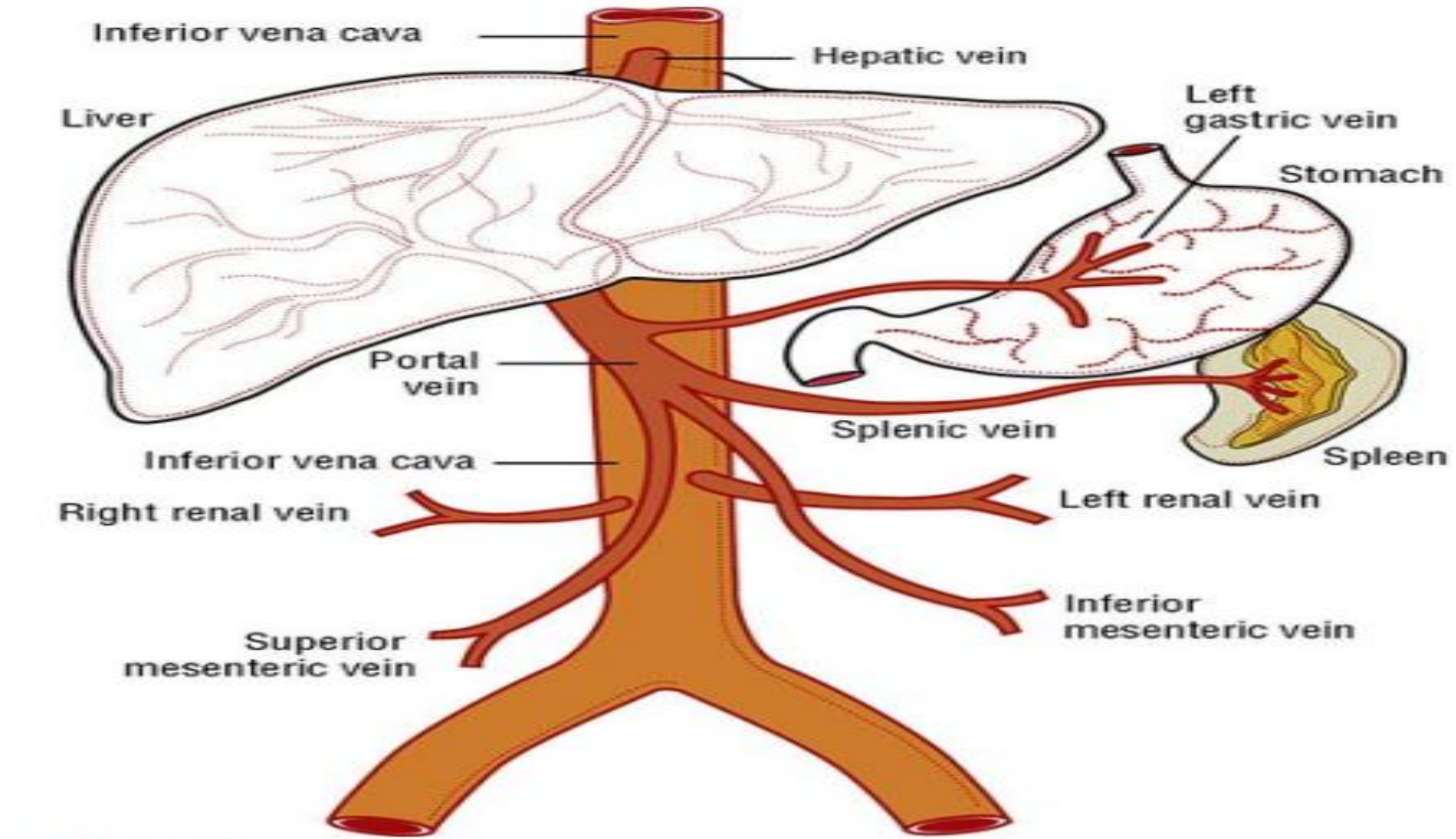
Epidemiology

- Cirrhosis is a leading cause of mortality and morbidity across the world.
- It is the 11th leading cause of death and 15th leading cause of morbidity.
- In 2019, cirrhosis accounted for 2.4% of global deaths, with a projected increase in deaths expected over the next decade.
- While the overall burden of cirrhosis remains higher in males, the increasing incidence among females and the changing etiological landscape is emerging.
- Acute variceal bleeding and spontaneous bacterial peritonitis (SBP) are among the immediately life-threatening complications of cirrhosis.
- Associated conditions causing significant morbidity include ascites and hepatic encephalopathy (HE).
- Approximately 50% of patients with cirrhosis develop ascites during 10 years of observation.
- Within 2 years, nearly half of patients who develop ascites will die.

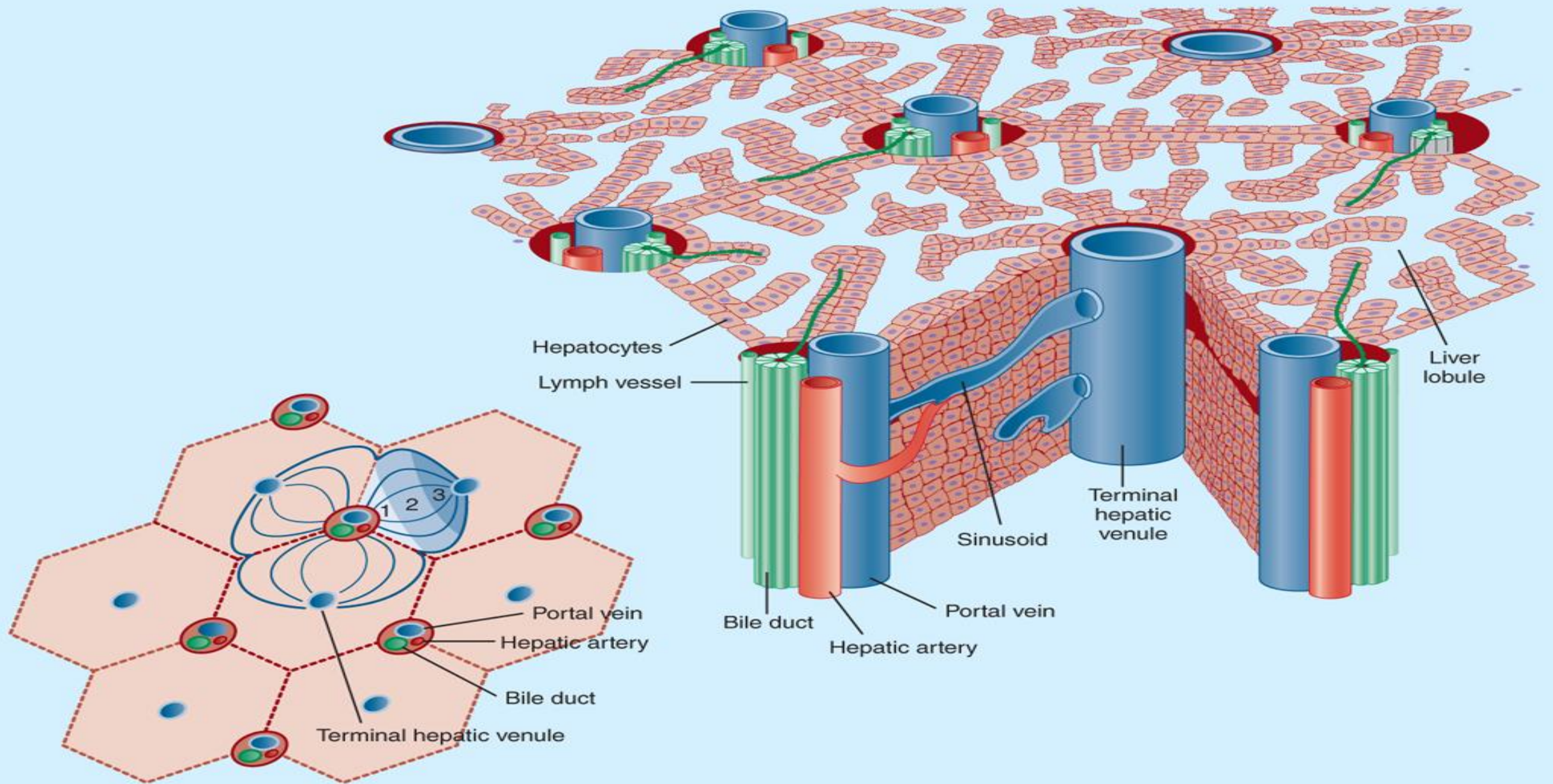
Etiology of Cirrhosis

- **Chronic alcohol consumption**
- **Chronic viral hepatitis (B& C)**
- **Metabolic liver disease**
 - ✓ Hemochromatosis
 - ✓ Wilson disease
 - ✓ α 1-Antitrypsin deficiency
 - ✓ Fatty liver
- **Immunologic disease**
 - ✓ Autoimmune hepatitis.
 - ✓ Primary biliary cholangitis (Primary biliary cirrhosis)
- **Vascular disease**
 - ✓ Budd–Chiari syndrome
 - ✓ Cardiac failure
- **Drugs**
 - ✓ Isoniazid, methyldopa, amiodarone, amoxicillin-clavulanate, nitrofurantoin, diclofenac, methotrexate, nevirapine, propylthiouracil, valproate

Pathophysiology



The portal venous system.



Continued...

- The liver can be thought of as an **elaborate blood filtration system** receiving blood from the **hepatic artery and the portal vein**.
- Portal blood originates from the mesenteric, gastric, splenic, and pancreatic veins.
- Blood enters the liver via the portal triad which contains branches of the portal vein, hepatic artery and bile ducts
- It then drains through the sinusoidal spaces of the hepatic lobule which are lined by the workhorses of the liver, the hepatocytes

Continued...

- When hepatic injury occurs, stellate cells in the affected areas begin to resemble fibroblasts,
 - Become a major source of collagen and other matrix proteins
- Eventually causing the permanent hepatic scarring characteristic of cirrhosis.
- The progressive deposition of fibrous material within the sinusoids disrupts the normal blood flow.
 - Decrease in endogenous vasodilators
 - Increase in endogenous vasoconstrictors Lead to complications

Complications of cirrhosis

- Cirrhosis and the pathophysiologic abnormalities result in:
 - Ascites
 - Portal Hypertension
 - Esophageal varices
 - Hepatic encephalopathy(HE)
 - Coagulation disorders
 - Malnutrition

Clinical Presentations

➤ Ranges from **asymptomatic**, with abnormal laboratory or radiographic tests, to **decompensated** with ascites, SBP, HE, or variceal bleeding.

➤ **Signs and Symptoms**

- Anorexia
- Malaise
- Weight loss
- Pruritus
- **Jaundice**
- Palmar erythema
- **Spider angiomata**
- Hyperpigmentation
- **Hepatomegaly**
- **Splenomegaly**
- **Ascites**
- **Edema**
- Pleural effusion
- Respiratory difficulties
- **Coagulopathy**
- **Mental status changes**

Clinical presentations.....

Laboratory Tests

- Hypoalbuminemia
- Elevated prothrombin time (PT)
- Thrombocytopenia
- Elevated AST, ALT, ALP and GGT
- Elevated Bilirubin

Diagnosis

- Complete Hx and P/E
- Lab evaluation
 - ✓AST, ALT, HCV, HBV
- Ratio of serum albumin & ascitic fluid albumin gradient(SAAG) level
 - >1.1 g/dL.....cirrhosis (ascites due pHTN)
 - <1.1 g/dL.....infectious or malignant cause of ascites
- Electrolytes
- CBC and **Ascitic fluid culture** (Platelet, bacterial peritonitis)
- Culture (when indicated) of blood or other specimens
- Imaging CT, MRI, EEG, US, Endoscopy
- Liver biopsy(staging)

Criteria and Scoring for the Child-Pugh Grading of Chronic Liver Disease

Score	1	2	3
Total bilirubin (mg/dL)	<2 (34.2 µmol/L)	2-3 (34.2- 51.3 µmol/L)	>3 (51.3 µmol/L)
Albumin (g/dL)	>3.5 (35 g/L)	2.8-3.5 (28-35 g/L)	<2.8 (28 g/L)
Ascites	None	Mild	Moderate
Encephalopathy (grade)	None	1 and 2	3 and 4
Prothrombin time (seconds prolonged)	<4	4-6	>6

Grade A, <7 points; grade B, 7-9 points; grade C, 10-15 points.

Data from Reference 15.

Child–Pugh classification system

- to assess and define severity of cirrhosis
- predict patient survival, surgical outcome,
- Risk of variceal bleeding.

Treatment

- ❑ **Dependent on complication(s)**

- ❑ Goal of therapy

- Resolution of acute complications.
- Tamponade of bleeding.
- Resolution of hemodynamic instability.
- Prevention of complications.
- Adequate lowering of portal pressure with medical therapy using nonselective β -adrenergic blocker therapy.
- Supporting abstinence from alcohol.

General Approach

- Identify and eliminate **causes of cirrhosis** (e.g., [alcohol](#) abuse).
- Evaluate for clinical signs of **ascites** and manage with pharmacologic treatment (e.g., **diuretics**) and **paracentesis**.
- Careful monitoring for spontaneous bacterial peritonitis (**SBP**) in patients with ascites who undergo acute deterioration.

General Approach.....

- **Assess risk for variceal bleeding** and begin pharmacologic prophylaxis where indicated, reserving endoscopic therapy for high-risk patients or acute bleeding episodes.
- HE is common complication of cirrhosis and requires clinical vigilance and treatment with dietary restriction, elimination of CNS depressants, and therapy to lower ammonia levels.
- Monitor for signs of hepatorenal syndrome, pulmonary insufficiency, and endocrine dysfunction.

Treatment of Complications

Ascites

- The pathologic accumulation of lymph fluid within the peritoneal cavity.
- It is one of the earliest and most common presentations of cirrhosis.
- Two possible proposed mechanisms
 - Portal HTN and hepatic insufficiency lead to splanchnic arterial vasodilation and decreased peripheral resistance b/c of vasodilators release (e.g. Nitric Oxide (NO)) →→ Systemic hypotension →→ ++ RAAS activation →→ Na⁺ and H₂O retention →→ Ascites
 - Decrease in serum Albumin level →→ increases Hydrostatic pressure ...leading to excessive splanchnic lymph production and leakage →→ The leaked fluid accumulates in the abdominal cavity →→ ascites

Ascites



Ascites & Spontaneous Bacterial Peritonitis (SBP)

- **Ascites:** accumulation of fluid within the peritoneal cavity
 - It is the most common complication of cirrhosis
- **Spontaneous bacterial peritonitis (SBP)**
 - *An infection of ascitic fluid* without evidence for an intra-abdominal secondary source such as a perforated viscus.
 - SBP is almost always seen in the setting of end-stage liver disease

Ascites Treatment

Goals of therapy

- To mobilize ascitic fluid
- To diminish abdominal discomfort, back pain & difficulty in ambulation
- To prevent major complications

Non-pharmacological management

- Na⁺ and water restriction, therapeutic paracentesis

Pharmacological management

- Diuretics
 - ✓ Spironolactone + Furosemide or Spironolactone alone

Continued....

- **Diuretics Therapy:** Choice of agent
 - High level of circulating aldosterone
 - Decrease excretion and increase production
 - Activation of RAAS
 - Hepatic impairment prolongs the half life of aldosterone
 - Low concentration of albumin
- Spironolactone is rational choice
 - Dose 100 mg to 200 mg up to 400 mg
- Combination with other diuretics: Furosemide
 - Spironolactone to Furosemide ratio (100: 40 mg)

Continued....

SBP

- >92% of all cases of SBP are monomicrobial, with
 - E. coli (most common), Klebsiella species, Other Gm(-) bacteria
 - Gm(+) organisms, Streptococcal (25%)
 - Anaerobic infection (rare <5%)
- Patients with documented or suspected SBP ...broad-spectrum antibiotic therapy
 - Third-generation cephalosporin.....the drug of choice for SBP
 - Ceftriaxone, 1gm, IV, BID for 7-10 days
 - Cefotaxime 2 gm IV every 8 hours or a similar third-generation cephalosporin **for 5 days**
 - Alternative: Ciprofloxacin, 200mg, IV, BID for 7-10 days
 - Oral ofloxacin 400 mg every 12 hours for **8 days**
 - ✓ an alternative for patients without vomiting, shock, significant HE, or serum creatinine >3 mg/dL.

SBP continued....

□ Prophylaxis against SBP

❖ Indications to start prophylaxis

- Patients who recover from an episode of SBP
 - Advanced cirrhosis and ascitic fluid protein lower than 1.5 g/dl without prior SBP
 - In patients with gastrointestinal bleeding and severe liver disease
- First line: Norfloxacin, 400mg, P.O. daily
 - Ciprofloxacin 500 mg or double-strength trimethoprim–sulfamethoxazole daily
 - Duration of prophylaxis: Generally indefinite but if there is long-lasting improvement with disappearance of ascites, prophylaxis can be discontinued.

Portal Hypertension

- Portal hypertension is characterized by prolonged elevation of the portal venous pressure
- Normal portal vein pressure is 5 to 10 mm Hg.
- Patients developing clinical features or complications of portal hypertension usually have portal venous pressures > 10 to 12 mmHg.
- Portal venous pressure is determined by the portal blood flow and portal vascular resistance.
 - Increased vascular resistance is usually the main factor producing portal HTN, irrespective of its cause.

- Schistosomiasis and cirrhosis are the most common cause of portal HTN world-wide
- Increased portal vascular resistance → gradual reduction in the flow of portal blood to the liver → development of collateral vessels (portal blood to bypass the liver and enter the systemic circulation directly).
- Collateral vessel formation is widespread but occurs particularly in the GIT (esophagus, stomach and rectum & in the anterior abdominal wall).
- Normally all the portal blood flows through the liver but, as collateral vessel formation progresses, half or more (and occasionally almost all) of the portal blood flow can be shunted directly to the systemic circulation.

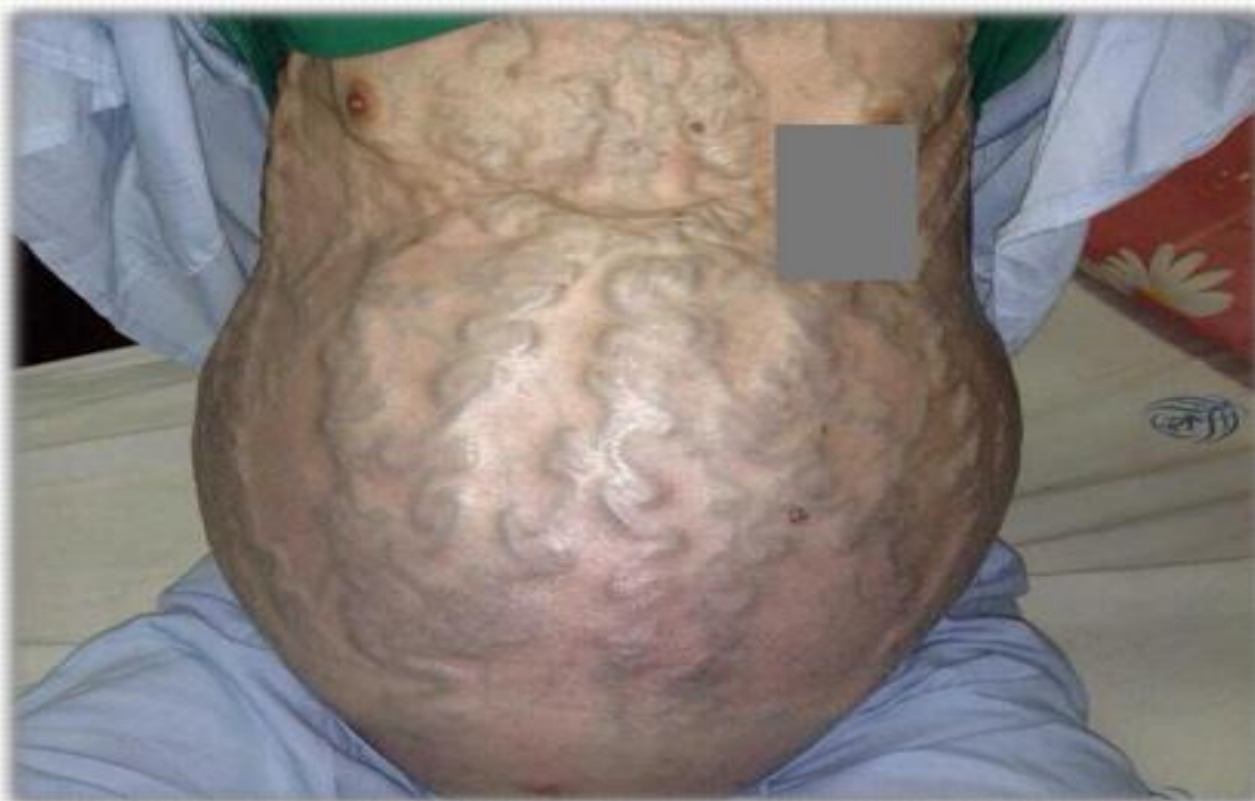
Clinical features portal HTN

- The clinical features of portal HTN results principally from portal venous congestion and collateral vessel formation.
- Splenomegaly is a cardinal finding,
 - Dx of portal HTN is unlikely when splenomegaly cannot be detected clinically or by ultrasonography.
- Hypersplenism (destruction of blood cell) is common →→ thrombocytopenia, anemia, leukopenia

Continued...

- Collateral vessels may be visible on the anterior abdominal wall and may form a **caput medusae**.
 - ❖ The most important collateral vessels occur in the esophagus and stomach, where they can cause severe bleeding.
 - ❖ Porto-systemic shunting of blood may contribute for hepatic encephalopathy of advanced liver disease.

Caput Medusae



Continued....

- Treatment of portal hypertension refers to management of its complications and correction of the cause whenever possible.
- Gastroesophageal varices & upper GI bleeding.
 - Prevention of bleeding
 - Acute management of bleeding
- Ascites management and peripheral edema –salt restriction, diuretic....

Portal HTN and Varices

- **Varices:** Alternative routes of blood flow from the portal to the systemic circulation, bypassing the liver
- Varices decompress the portal venous system and return blood to the systemic circulation.
 - Risk of varices high, when portal pressure exceed the vena cava pressure by > 10 mmHg
- Occur at any level of the GIT
- Varices are weak superficial vessels; any additional increase in pressure can cause them to rupture and bleed

Continued

How Varices formed??

- Portal hypertension causes blood flow to be forced backward, causing veins to enlarge and varices to develop across the esophagus and stomach from the pressure in the portal vein.
- The backup of pressure also causes the spleen to become enlarged

Continued...

- The management of varices involves three strategies:
 - Primary prophylaxis: prevention of the first bleeding episode
 - Treatment of acute variceal hemorrhage
 - Secondary prophylaxis: prevention of rebleeding (Hx of previously bled)

Primary Prophylaxis

❑ Beta -Adrenergic Blockade

- Mainstay: **nonselective β -adrenergic blocking agents** (propranolol 10mg PO TID or nadolol 20mg QD)
 - Reduce portal pressure by reducing portal venous inflow via two mechanisms:
 - ✓ Decrease in CO (β_1) and splanchnic blood flow (β_2)
- Propranolol average dose of (60 mg/d)
- HR goal: **55 to 60 beats/min** & Systolic BP not less than 90 mm Hg
- Nitrates may be considered, if the patient is intolerant or C/I to B-blockers

Acute Variceal Hemorrhage

- An emergency and feared condition of cirrhosis
- Initial treatment goals include:
 - ✓ Replenishment of blood volume and correction of coagulopathy must be done with packed RBC (to increase Hgb 7 to 9 g/dL) and fresh frozen plasma and platelets
 - ✓ Protection of airway from aspiration of blood
 - ✓ Antibiotics to prevent SBP and Gm(-) systemic infection
 - IV ceftriaxone 1 g/24 hours or oral ciprofloxacin 500 mg twice daily, or trimethoprim–sulfamethoxazole one double-strength tablet twice daily
 - ✓ Control of bleeding
 - ✓ Prevention of re-bleeding, and preservation of liver function

Continued....

Drug Therapy

- ✓ Vasoactive drug (somatostatin, **octreotide**, or terlipressin)
 - ✓ IV bolus of 50 to 100 mcg & followed by a continuous infusion of 25 mcg/h →→50 mcg/h.
 - ✓ Work as splanchnic **vasoconstrictors, thus decreasing portal blood flow and pressure**
- Child–Pugh Class C and B patients with active hemorrhage at the time of diagnostic endoscopy are at high risk for failing standard therapy. Hence, Endoscopic Therapy may be indicated
 - ✓ Endoscopic variceal band ligation (EVL)
 - ✓ Surgery (Transjugular intrahepatic portosystemic shunt (TIPS))
- **Secondary Prophylaxis**: Prevention of re-bleeding
 - Combination therapy with **beta-adrenergic blockers and chronic EVL**

Hepatic Encephalopathy (HE)

- CNS disturbance with a wide range of neuro-psychiatric symptoms
- Gut-derived neurotoxins that are not removed by the liver due to vascular shunting and decreased hepatic mass →→get to the brain and cause the symptoms.
 - NH₃, glutamate and other metabolites
- These substances enter the CNS, they cause alterations of neurotransmission that affect consciousness and behavior.
- The hallmark of persistent HE is chronic **cognitive deficits**

Hepatic Encephalopathy (HE)...

- ❖ Hepatic encephalopathy is categorized as type A, B, or C.
- **Type A** is HE induced by acute liver failure
- **Type B** is due to portal-systemic bypass without associated intrinsic liver disease, and
- **Type C** is HE that occurs in patients with cirrhosis.

Grading System for Hepatic Encephalopathy

Grade	Level of Consciousness	Personality/Intellect	Neurologic Abnormalities
Unimpaired	Normal	Normal	Normal
Minimal	No clinical evidence of change	No clinical evidence of change/alterations identified on psychometric or neuropsychological testing	No clinical evidence of change
I	Trivial lack of awareness; shortened attention span	Euphoria or anxiety; impairment of addition or subtraction	Altered sleep rhythm
II	Lethargic	Obvious personality changes; inappropriate behavior; apathy	Asterixis; dyspraxia; disoriented for time
III	Somnolent but arousable	Bizarre behavior	Responsive to stimuli; confused; gross disorientation to time and space
IV	Coma/unarousable	None	Does not respond to stimuli

HE Treatment

- The principles of treatment are
 - ✓ To treat or remove precipitating causes
 - ✓ To reduce or eliminate protein intake and
 - ✓ To suppress production of neurotoxins by bacteria in the bowel
- The mainstay of therapy of HElower blood ammonia concentrations
 - ✓ diet therapy; limit protein intake to 1.2 to 1.5 g/kg/day.
 - ✓ lactulose, and
 - ✓ antibiotics alone or in combination with lactulose

Continued....

- ❖ Main stay is reducing NH₃ blood concentration
- In episodic HE, **lactulose** 25 ml (16.7 g) orally every 1 to 2 hours until catharsis begins and the patient experiences one to two bowel movements.
 - Then 15 to 45 mL orally every 8 to 12 hours and titrated to produce two to three soft stools per day for chronic therapy.
- **Rifaximin** 550 mg twice daily plus lactulose is superior to lactulose alone in patients with a history of recurrent HE.
- Because of its more favorable adverse effect profile, rifaximin is now considered the next line of therapy for recurrent HE over either metronidazole or neomycin.
- Metronidazole/neomycin is reserved for non respondents to diet and lactulose.
 - Inhibit urease-producing bacteria

Continued...

□ How lactulose works?????

- Enhances the net movement of ammonia from the bowel...laxative
- Lactulose also decreases survival of urease producing bacteria and promotes conversion of ammonia to ammonium which is not as readily absorbed in the gut.
- Lactulose is metabolized by gut flora into acetic acid and lactic acid, which lower colonic pH and create a cathartic effect to increase fecal nitrogen waste.
 - Acidification of the colon →→ decrease NH_3 ($\text{NH}_3 \rightarrow \text{NH}_4^+$)

Coagulation defects

- Coagulopathies signal end-stage liver disease.
- The liver manufactures both procoagulant and anticoagulant factors essential for blood clotting and maintenance of blood homeostasis.
- In advanced disease, the liver is unable to synthesize these proteins, resulting in extended clotting times (eg, prothrombin time) and bleeding
- **Thrombocytopenia** is common in advanced liver disease as a result of both **decreased hepatic thrombopoietin stimulation in the bone marrow** and **splenic sequestration of formed platelets**.
- The net effect of these events is the development of bleeding diathesis.

Coagulation defects

- Patients with cirrhosis who are actively bleeding receive resuscitation with packed red blood cells targeting a hemoglobin of 7 to 9 g/dL.
- Platelet transfusion is used to maintain platelets over 50,000/mm³ during the period of active bleeding.

Hepatorenal Syndrome

- Functional renal failure without renal pathology
- Portal hypertension → the release of vasodilators → splanchnic vasculature reducing renal blood flow causing renal hypo-perfusion.
- This causes activation of the RAAS → severe renal vasoconstriction and fluid retention..... leading to ascites.
- As liver disease progresses, systemic vasodilation worsens and, subsequently, increased renal vasoconstriction occurs and renal blood flow is further decreased.
- The result is hepatorenal syndrome that can be separated into two main types.
 - Type 1 occurs rapidly and has a precipitating cause such as SBP.
 - Type 2 occurs more slowly and does not have a precipitating event

Continued...

- Rx Systemic vasoconstrictors
 - ✓ Arterial vasoconstrictors such as Somatostatin analogues (octreotide), and alpha-adrenergic agonists (midodrine)
 - ✓ Avoid all potential nephrotoxins(aminoglycosides, diuretic, NSAIDs)
 - ✓ IV albumin at a dose of 1 g/kg up to a maximum of 100 g.
 - ✓ Dialysis
 - ✓ Liver transplantation →→refractory HRS

Malnutrition

- Liver is principally involved in the regulation of protein and energy metabolism
- It is not surprising that patients with advanced liver disease are commonly malnourished
- Once patients become cirrhotic, they are more catabolic, and muscle protein is metabolized.
- Multiple factors
 - ✓ Poor dietary intake
 - ✓ Alterations in gut nutrient absorption, and
 - ✓ Alterations in protein metabolism

Evaluation of therapeutic outcomes

- Ascites
 - Daily assessment of weight
 - Prevent or eliminate ascites and its secondary complications
- SBP
 - Evidence of clinical deterioration (e.g., abdominal pain, fever, anorexia, malaise, fatigue)
 - Prevent/treat infection to decrease mortality
- Variceal bleeding
 - Endoscopy, CBC, evidence of overt bleeding
 - Appropriate reduction in HR and portal pressure
- Monitor adverse effects of drugs

Drug Monitoring Guidelines

Drug	Adverse Drug Reaction	Monitoring Parameter	Comments
Nonselective β -adrenergic blocker	Heart failure, bronchospasm, glucose intolerance	BP, HR Goal HR: 55-60 beats/min or maximal tolerated dose	Nadolol, propranolol, carvedilol
Octreotide	Bradycardia, hypertension, arrhythmia, abdominal pain	BP, HR, EKG, abdominal pain	
Spironolactone/furosemide	Electrolyte disturbances, dehydration, renal insufficiency, hypotension	Serum electrolytes (especially potassium), SCr, blood urea nitrogen, BP Goal sodium excretion: >78 mmol/day	Spot urine sodium concentration greater than potassium concentration correlates well with daily sodium excretion >78 mmol/day
Lactulose	Electrolyte disturbances	Serum electrolytes Goal number of soft stools per day: 2-3	
Neomycin	Ototoxicity, nephrotoxicity	SCr, annual auditory monitoring	
Metronidazole	Neurotoxicity	Sensory and motor neuropathy	
Rifaximin	Nausea, diarrhea		

3P, blood pressure; HR, heart rate; beats/min, beats per minute; EKG, electrocardiogram; SCr, serum creatinine; mmol, millimole.

Thank You